MSMS 401 : Bayesian Statistics and Computation

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Problem

Let $X_i \overset{\text{iid}}{\sim} \text{Geometric}(\theta)$ on support $\{0, 1, 2, \ldots\} \forall i = 1(1)n$ and *a priori* $\theta \sim \mathscr{B}_1(\alpha, \beta)$, both α and β are unknown. Obtain empirical Bayes estimate of α and β.

Solution

$$f(x \mid \theta) = \theta(1-\theta)^x, \qquad x = 0, 1, 2, \ldots, \quad 0 < \theta < 1.$$

$$\pi(\theta; \alpha, \beta) = \frac{1}{\mathcal{B}(\alpha, \beta)} \cdot \theta^{\alpha-1}(1-\theta)^{\beta-1}, \qquad 0 < \theta < 1.$$

Then the marginal distribution of X will be

$$\begin{aligned}
m(x_i; \alpha, \beta) &= \int_0^1 f(x_i \mid \theta) \cdot \pi(\theta; \alpha, \beta)\, d\theta \\
&= \int_0^1 \theta(1-\theta)^{x_i} \cdot \frac{1}{\mathcal{B}(\alpha, \beta)} \cdot \theta^{\alpha-1}(1-\theta)^{\beta-1}\, d\theta \\
&= \frac{1}{\mathcal{B}(\alpha, \beta)} \cdot \int_0^1 \theta^{\alpha+1-1}(1-\theta)^{\beta+x_i-1}\, d\theta \\
&= \frac{1}{\mathcal{B}(\alpha, \beta)} \cdot \mathcal{B}(\alpha+1, \beta+x_i) \\
&= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+1)\Gamma(\beta+x_i)}{\Gamma(\alpha+\beta+x_i+1)} \\
&= \frac{\Gamma(\alpha+\beta) \cdot \Gamma(\alpha+1) \cdot \Gamma(x_i+\beta)}{\Gamma(\alpha) \cdot \Gamma(\beta) \cdot \Gamma(\alpha+\beta+x_i+1)}
\end{aligned}$$

For a sample of size n, the marginal likelihood will be

$$\begin{aligned} L(\alpha, \beta) &= \prod_{i=1}^{n} m(x_i; \alpha, \beta) \\ &= \prod_{i=1}^{n} \frac{\Gamma(\alpha+\beta)\Gamma(\alpha+1)\Gamma(x_i+\beta)}{\Gamma(\alpha)\Gamma(\beta)\Gamma(\alpha+\beta+x_i+1)}. \end{aligned}$$

The log-likelihood will be

$$\begin{aligned} \ell(\alpha, \beta) = n \ln\Gamma(\alpha+\beta) + n \ln\Gamma(\alpha+1) + \sum_{i=1}^{n} \ln\Gamma(x_i+\beta) \\ - n \ln\Gamma(\alpha) - n \ln\Gamma(\beta) - \sum_{i=1}^{n} \ln\Gamma(\alpha+\beta+x_i+1). \end{aligned}$$

Here both α and β are unknown. Partial differentiation of $\ell(\alpha, \beta)$ w.r.t. α and β introduces **digamma** function. Hence, analytical solution of the equations

$$\begin{aligned} \frac{\partial}{\partial\alpha}\ell(\alpha, \beta) = 0 \\ \frac{\partial}{\partial\beta}\ell(\alpha, \beta) = 0 \end{aligned}$$

is not possible and we resort to numerical methods for approximate solutions.